# EAS439

**Timed Online Assignment – January Semester 2022**

## Numerical Analysis

**Tuesday, 17 May 2022** **1:00 pm – 3:00 pm**

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**Time allowed: 2 hours**

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**INSTRUCTIONS TO STUDENTS:**

1. This Timed Online Assignment (TOA) contains **FOUR (4)** questions and comprises **SEVEN (7)** pages (including cover page).

2. You must answer **ALL** questions.

3. If you have any queries about a question, or believe there is an error in the question, briefly explain your understanding and assumptions about that question before attempting it.

4. You MUST **submit your answers via Canvas (similar to TMA submission) at the end time of this TOA** (as stated on this cover page). The 15 minutes grace period as shown on Canvas is strictly meant for technical issues encountered during submission. Thereafter, you will not be able to submit your answers and you will be considered as having withdrawn from the course. **No appeal will be allowed.**

5. You are to include the following particulars in your submission: TOACourse Code, Your Name and SUSS PI No.. Name your submission file as TOAEAS439_FullName_StudentPI.

6. Your submission should consist of only one file and **must not exceed 500MB** in size. The file can be a Microsoft Word file saved in .doc or .docx format or a PDF file generated from scanned or photographed images.

7. All answers can be typed or hand written. In a Word file, flowcharts and graphs may be scanned or photographed and embedded provided it does not exceed the file size limit of 500MB.

8. All uploaded hand written answers must be clear, readable and complete. Marks will not be awarded for un-readable or incomplete images. Please ensure that question number is clearly stated on each page.

9. To prevent plagiarism and collusion, your submission will be reviewed by Turnitin and marker. The Turnitin report will only be made available to the marker and you will not be able to view it. Your Turnitin report will be examined thoroughly as part of the marking process.

10. The University takes a tough stance against plagiarism and collusion. Serious cases will normally result in the student being referred to SUSS's Student Disciplinary Group. For other cases, significant marking penalties or expulsion from the course will be imposed.

*Answer all questions. (Total 100 marks)*

**Question 1**

(a) Figure Q1(a) shows the forces that is acting on the aircraft in flight.

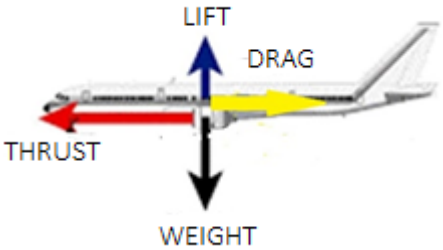


**Figure Q1(a)**

From Newton's 2nd law, the governing equation is:

$$m\frac{dV}{dt} = T - D$$

Where $V$ is the aircraft velocity.

(i) Formulate numerical approximation equations using *forward difference* to compute the aircraft speed $V$ during an acceleration.

(8 marks)

(ii) A turbo-fan aircraft has the following characteristics:

Mass : $m = 100{,}000\ kg$
Lift: $L = 38.22\ V^2$
Drag: $D = 2.327\ V^2$
Thrust : $T = 90{,}000\ N$
Initial Speed : $V = 100\ m/s$

Based on the numerical approximation formulated in Q1(a)(i), calculate the aircraft speed $V$ over time during level acceleration, using $dt = 4$. Copy Table Q1(a)(ii) to answer the question.

| Time (s) | aircraft speed $V$ (m/s) |
|---|---|
| 4 | |
| 8 | |
| 12 | |

**Table Q1(a)(ii)**

(8 marks)

(iii) Analyse the effect of $dt$ on the accuracy and efficiency of the numerical formulation used.

(4 marks)

(b) Analyse the truncation error of the numerical method formulated in Q1(a). What is the difference between the "Truncation Error" and the "Total Error"?

(5 marks)

**Question 2**

(a) Given the following equations:

$$4x + 2y + z = 7$$
$$0.2z - 5y = 0$$
$$2y + 0.1x - 2 = 6z$$

Using initial guess provided, calculate the solution by Gauss Seidel Iteration method. Using the table below, show all steps of the computation. Copy Table Q2(a) to answer the question.

| Iteration | x | y | z |
|---|---|---|---|
| Initial guess | 0 | 0 | 0 |
| 1 | | | |
| 2 | | | |
| 3 | | | |
| 4 | | | |
| 5 | | | |

**Table Q2(a)**

(9 marks)

(b) An aircraft flying at altitude of 5000 m, the power required for the aircraft as a function of cruising speed is given by:

$$P_R = 3.2345\, V^3 - 1050.2\, V^2 + 130420\, V + 465.23$$

The engine power set at part power is given by the equation:

$$P_A = 125500\, V - 387\, V^2$$

Use the initial guess: $V = 160\, m/s,$ formulate a computation procedure to determine the operating speed of the aircraft using Newton's Raphson method. The accuracy should be up to 1 decimal point in m/s.

(8 marks)

(c) The rate of climb (in fpm : feet per minute) of an aircraft is given by the following equation:

$$\frac{dh}{dt} = 1500e^{-0.055(h/1000)}$$

Use Simpson's 3/8 rule to formulate procedure to compute the time taken for the aircraft to climb from 1000 ft to 10000 ft.

(8 marks)

**Question 3**

(a) Consider the 1-D heat diffusion equation:

$$\frac{\partial T}{\partial t} = k\frac{\partial^2 T}{\partial x^2}$$

(i) Derive the equation into a convective equation of heat flux: $q = -k\frac{\partial T}{\partial x}$

(3 marks)

(ii) Formulate a numerical scheme using finite volume and explicit in time for the derived convective equation. Indicate clearly the flux for each finite volume.

(5 marks)

(b) Consider a thin rod is insulated at all point except at both ends as shown in Figure Q3(b)-1:

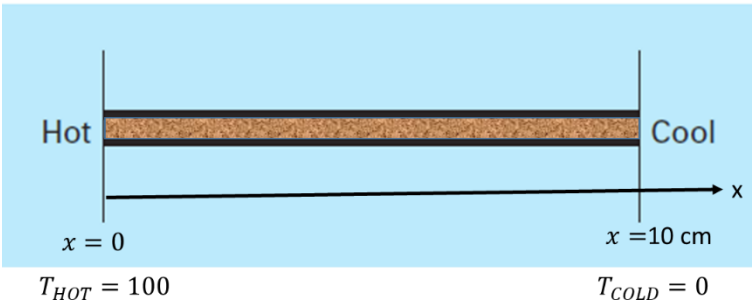


**Figure Q3(b)-1**

The initial temperature of the rod is $T(x) = 0$, and $k = 0.835\ cm^2/s$. A numerical scheme is designed by discretising the rod into FIVE finite volume, with the control point at each volume center as shown in Figure Q3(b)-2:

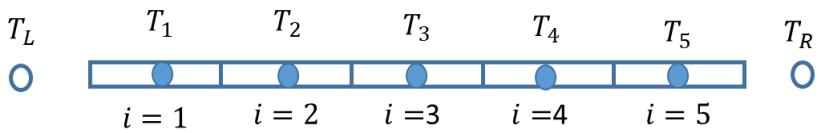


**Figure Q3(b)-2**

Where $T_L$ and $T_R$ are used to define boundary conditions at Hot end and Cold end respectively.

(i) Calculate the "Left flux" and "Right flux" for each finite volume. Copy Table Q3(b)(i) to answer the question.

| Volume | Left Flux $q_L$ | Right Flux $q_R$ |
|---|---|---|
| 1 | | |
| 2 | | |
| 3 | | |
| 4 | | |
| 5 | | |

**Table Q3(b)(i)**

(6 marks)

(ii) In an iteration using $\Delta t = 0.1$: at $t = 3$, the temperatures are as follow:

| $T_L$ | $T_1$ | $T_2$ | $T_3$ | $T_4$ | $T_5$ | $T_R$ |
|---|---|---|---|---|---|---|
| 100 | 41.00079 | 11.61641 | 3.479259 | 6.077958 | 20.53331 | 50 |

Compute T1 to T5 for $t = 3.1$.

(6 marks)

(iii) What is the suitable choice of $\Delta t$ for the scheme to be stable?

(2 marks)

(c) Appraise the difference between the Finite Volume method as compared to a Finite difference method.

(3 marks)

**Question 4**

(a) The distribution acting on the wing of the aircraft can be approximated by a distributed load acting on the main spar of the wing as shown in Figure Q4(a):

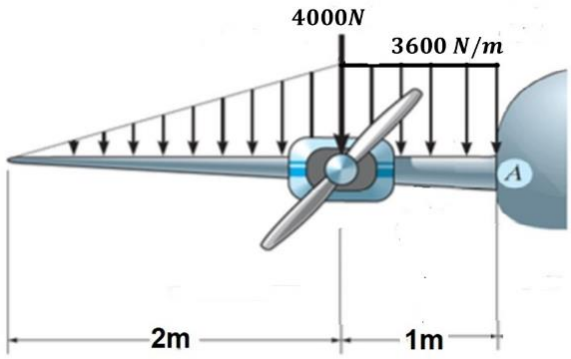


**Figure Q4(a)**

Formulate a finite element model to analyse the loading of the aircraft by using THREE beam elements. This model is to be used to appraise the shear force and bending moment that is acting on the main spar at the wing root (i.e. Point A).

Where the element stiffness matrix for beam element is:

$$\begin{Bmatrix} f_{1y} \\ m_1 \\ f_{2y} \\ m_2 \end{Bmatrix} = \frac{EI}{L^3} \begin{bmatrix} 12 & 6L & -12 & 6L \\ 6L & 4L^2 & -6L & 2L^2 \\ -12 & -6L & 12 & -6L \\ 6L & 2L^2 & -6L & 4L^2 \end{bmatrix} \begin{Bmatrix} v_1 \\ \phi_1 \\ v_2 \\ \phi_2 \end{Bmatrix}$$

Assume that the spar has a constant cross section where $EI$ is constant

(i) Construct the stiffness matrix for each of the beam element.

(4 marks)

(ii) Develop and compose the stiffness matrix of the entire wing spar.

(8 marks)

(b) Appraise the equivalent load at each nodal point for the solution of the finite element model.

(10 marks)

(c) What are the possible choice of numerical method that could be employed or design to compute the equation?

(3 marks)

**----- END OF PAPER -----**